Combinational and Arithmetic Circuits

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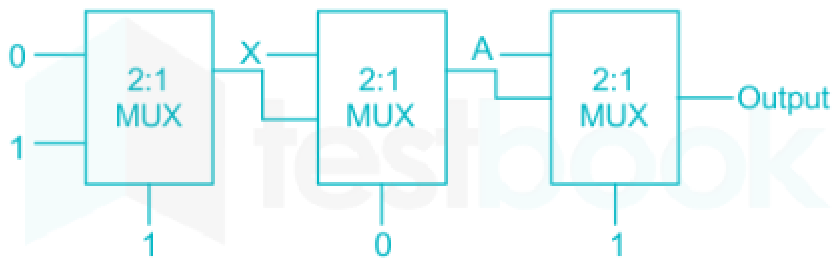
The output of the circuit shown is Fig is _______



1. 0
2. X
3. A
4. 1

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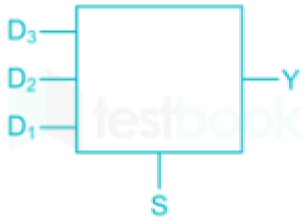
The combinational circuit shown in the below figure has the function. When selector input 's' is high the circuit is to detect if one of the data lines has logic '1' and no more than one data line is having logic '1', when selector input 'S' is low, the circuit will output '0', regardless of what is on the data lines. The logic of output Y is



1. $S(D_1 + D_2 + D_3)$
2. $S\left(\bar{D}_1 + D_2 + D_3\right)$
3. $\bar{S}\,(D_1 + D_2) + SD_3$
4. $(D_1 + D_3) + \bar{S}\,D_2$

S	D_3	D_2	D_1	Y
0	x	x	x	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	1
1	0	1	1	x
1	1	0	1	x

SD_3 \ D_2D_1	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	x	x	x
10	0	1	x	1

$Y = SD_3 + SD_1 + SD_2$

2. Digital Logic and Microprocessor

Er. Pralhad Chapagain

Syllabus

2.1 Digital logic: Number Systems, Logic Levels, Logic Gates, Boolean algebra, Sum-of-Products Method, Product-of-Sums Method, Truth Table to Karnaugh Map. (AExE0201)

2.2 Combinational and arithmetic circuits: Multiplexetures, Demultiplexetures, Decoder, Encoder, Binary Addition, Binary Subtraction, operation on Unsigned and Signed Binary Numbers. (AExE0202)

2.3 Sequential logic circuit: RS Flip-Flops, Gated Flip-Flops, Edge Triggered Flip-Flops, Mater- Slave Flip-Flops. Types of Registers, Applications of Shift Registers, Asynchronous Counters, Synchronous Counters. (AExE0203)

2.4 Microprocessor: Internal Architecture and Features of microprocessor, Assembly Language Programming. (AExE0204)

2.5 Microprocessor system: Memory Device Classification and Hierarchy, Interfacing I/O and Memory Parallel Interface. Introduction to Programmable Peripheral Interface (PPI), Serial Interface, Synchronous and Asynchronous Transmission, Serial Interface Standards. Introduction to Direct Memory Access (DMA) and DMA Controllers. (AExE0205)

2.6 Interrupt operations: Interrupt, Interrupt Service Routine, and Interrupt Processing. (AExE0206)

2.3 Sequential logic circuit: RS Flip-Flops, Gated Flip-Flops, Edge Triggered Flip-Flops, Mater- Slave Flip-Flops. Types of Registers, Applications of Shift Registers, Asynchronous Counters, Synchronous Counters. (AExE0203)

SEQUENTIAL LOGIC

- Sequential circuit produces an output based on current input and previous input variables.
- That means sequential circuits include memory elements which are capable of storing binary information.
- That binary information defines the state of the sequential circuit at that time.
- A latch is capable of storing one bit of information.
- As shown in figure there are two types of input to the combinational logic :
  - External inputs which not controlled by the circuit.
  - Internal inputs which are a function of a previous output states.

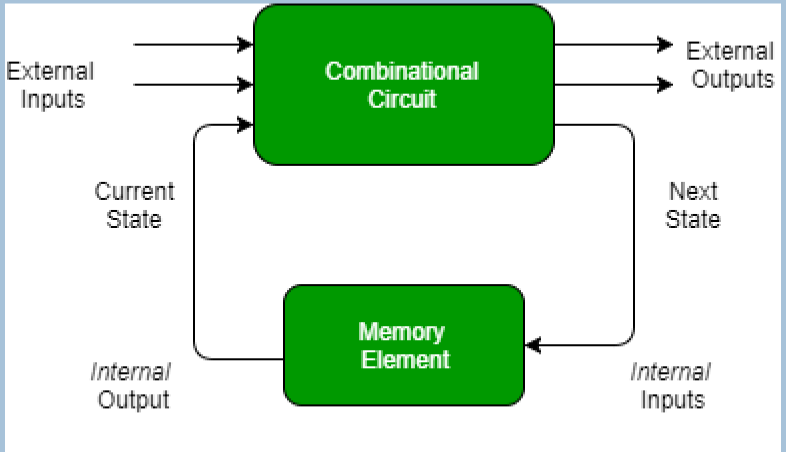


Figure: Sequential Circuit

SEQUENTIAL LOGIC - TYPES

Asynchronous sequential circuit

- These circuit do not use a clock signal but uses the pulses of the inputs.
- These circuits are faster than synchronous sequential circuits because they change their state immediately when there is a change in the input signal.
- We use asynchronous sequential circuits when speed of operation is important and independent of internal clock pulse.
- But these circuits are more difficult to design and their output is uncertain.

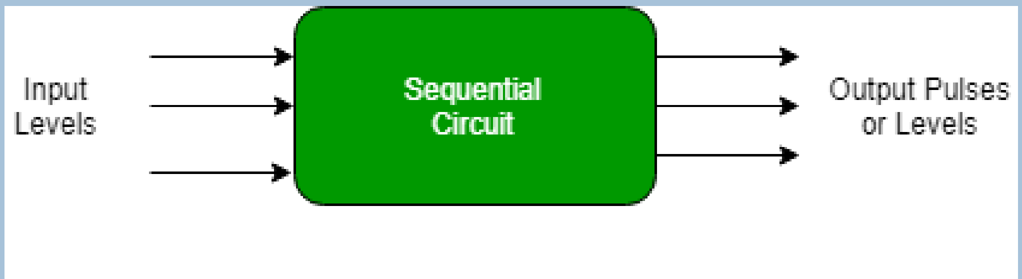


Figure: Asynchronous Sequential Circuit

SEQUENTIAL LOGIC - TYPES

Synchronous sequential circuit

- These circuit uses clock signal and level inputs (or pulsed) with restrictions on pulse width and circuit propagation.
- The output pulse is the same duration as the clock pulse for the clocked sequential circuits.
- Since they wait for the next clock pulse to arrive to perform the next operation, so these circuits are bit slower compared to asynchronous.
- Level output changes state at the start of an input pulse and remains in that until the next input or clock pulse.
- We use synchronous sequential circuit in synchronous counters, flip flops, and in the design of state management machines.

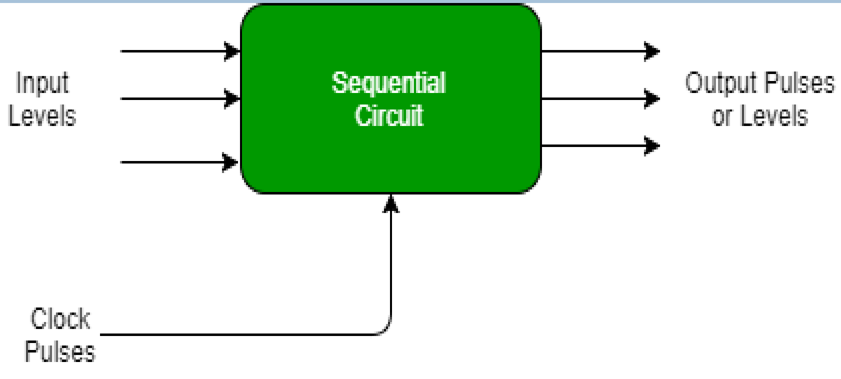
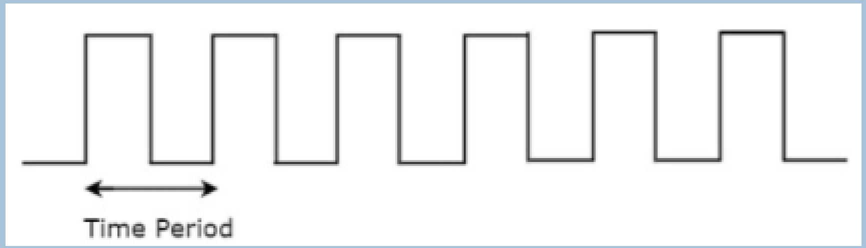


Figure: Synchronous Sequential Circuit

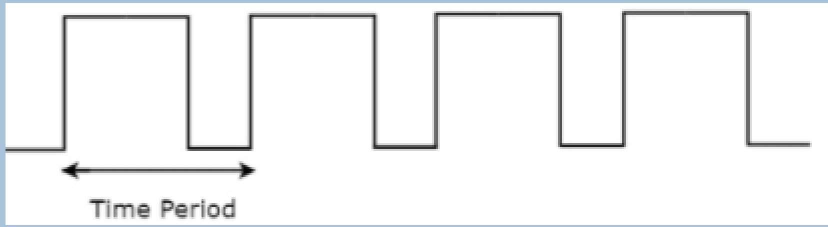
CLOCK SIGNAL AND TRIGGERING

Clock signal:

- Clock signal is a periodic signal and its ON time and OFF time need not be the same.
- We can represent the clock signal as a square wave, when both its ON time and OFF time are same.
- The pattern repeats with some time period. In this case, the time period will be equal to either twice of ON time or twice of OFF time.



- We can represent the clock signal as train of pulses, when ON time and OFF time are not same
- In this case, the time period will be equal to sum of ON time and OFF time.



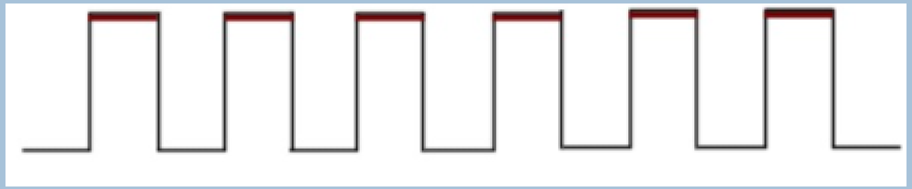
CLOCK SIGNAL AND TRIGGERING

Types of Triggering

- Level triggering
- Edge triggering

Level triggering

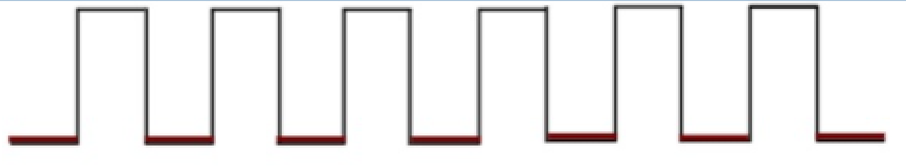
- There are two levels, namely logic High and logic Low in clock signal.
- Following are the two types of level triggering.
  - Positive level triggering
  - Negative level triggering



- If the sequential circuit is operated with the clock signal when it is in Logic High, then that type of triggering is known as **Positive level triggering**. It is highlighted in above figure.

CLOCK SIGNAL AND TRIGGERING

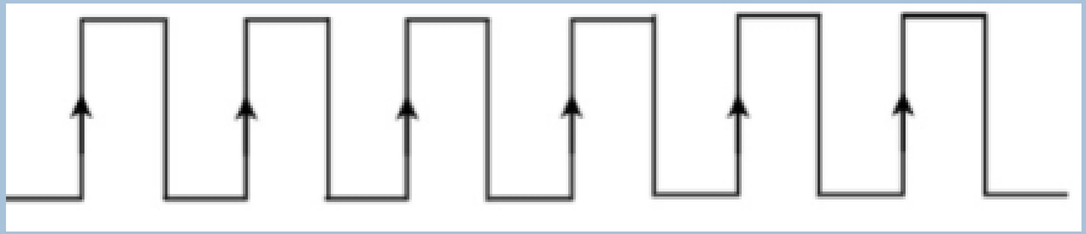
If the sequential circuit is operated with the clock signal when it is in Logic Low, then that type of triggering is known as **Negative level triggering**. It is highlighted in the following figure.



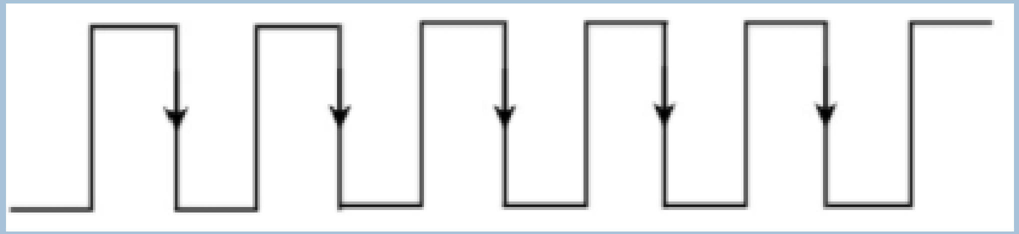
Edge triggering

- There are two types of transitions that occur in clock signal. That means, the clock signal transitions either from Logic Low to Logic High or Logic High to Logic Low..
  - Positive edge triggering
  - Negative edge triggering
- If the sequential circuit is operated with the clock signal that is transitioning from Logic Low to Logic High, then that type of triggering is known as **Positive edge triggering**. It is also called as rising edge triggering. It is shown in the following figure.

CLOCK SIGNAL AND TRIGGERING



- If the sequential circuit is operated with the clock signal that is transitioning from Logic High to Logic Low, then that type of triggering is known as **Negative edge triggering**. It is also called as falling edge triggering. It is shown in the following figure.



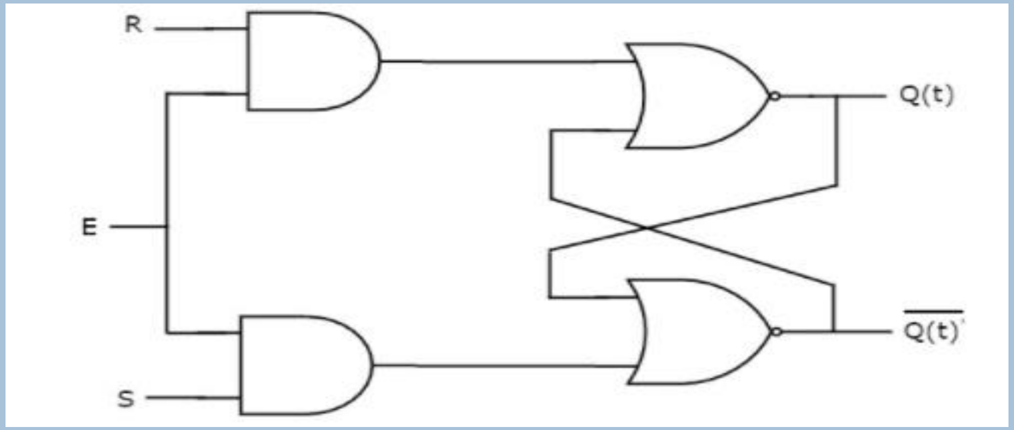
LATCHES

- There are two types of memory elements based on the type of triggering that is suitable to operate it.
  - Latches
  - Flip-flops
- Latches operate with enable signal, which is level sensitive. Whereas, flip-flops are edge sensitive.

SR Latch

- SR Latch is also called as Set Reset Latch.
- This latch affects the outputs as long as the enable, E is maintained at '1'. The circuit diagram of SR Latch is shown in the following figure

S	R	Q $t+1$
0	0	Q t
0	1	0
1	0	1
1	1	-

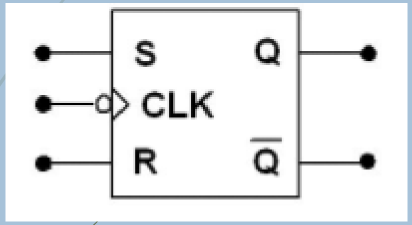


FLIP-FLOP

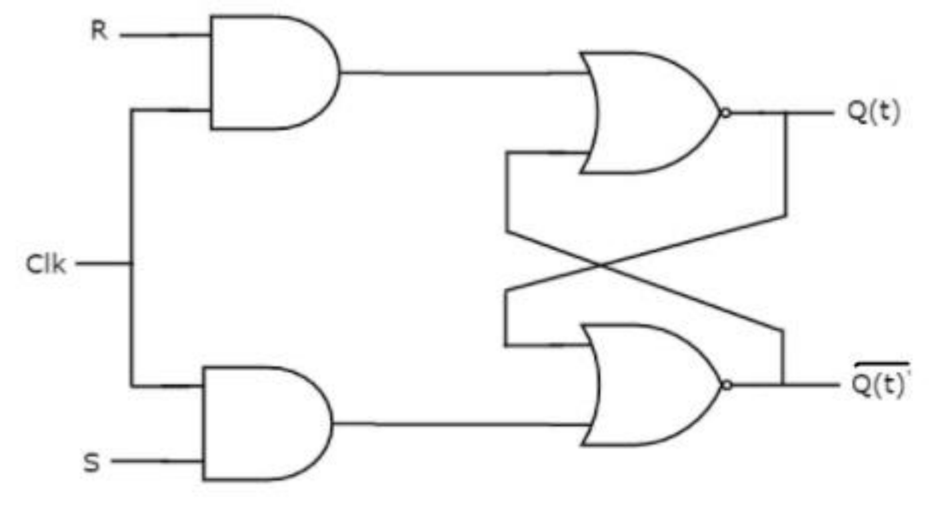
- Flip-flop is a basic digital memory circuit, which stores one bit of information.
- Flip flops are the fundamental blocks of most sequential circuits.
- Flip-flops are used as memory elements in clocked sequential circuit.
- Flip-flop circuit has two outputs, one for the normal value and one for the complement value of the bit stored in it.
- Binary information can be enter a flip-flop in a variety of ways, a fact, which gives rise to different types of flip-flops.
  - SR Flip-Flop
  - D Flip-Flop
  - JK Flip-Flop
  - T Flip-Flop

FLIP-FLOP – SR FLIP-FLOP

SR (Set-Reset) flip-flop operates with only positive clock transitions or negative clock transitions. Whereas, SR latch operates with enable signal. The circuit diagram of SR flip-flop is shown in the following figure.



Graphical Symbol



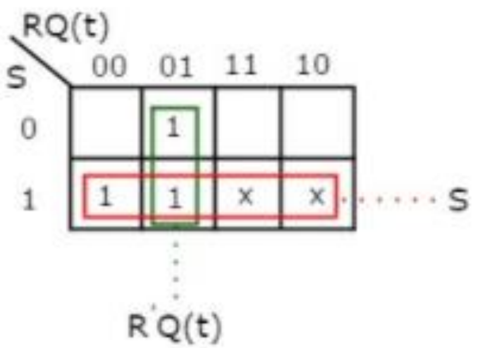
Logic diagram

Clock	S	R	Q(t)	Q(t+1)	Comment
↓	0	0	0	0	Hold
↓	0	0	1	1	Hold
↓	0	1	0	0	Reset
↓	0	1	1	0	Reset
↓	1	0	0	1	Set
↓	1	0	1	1	Set
↓	1	1	0	X	Indeterminant
↓	1	1	1	X	Indeterminant

Characteristic Table

FLIP-FLOP – SR FLIP-FLOP

K-Map for next state, Q $t + 1$ is shown in the following figure.

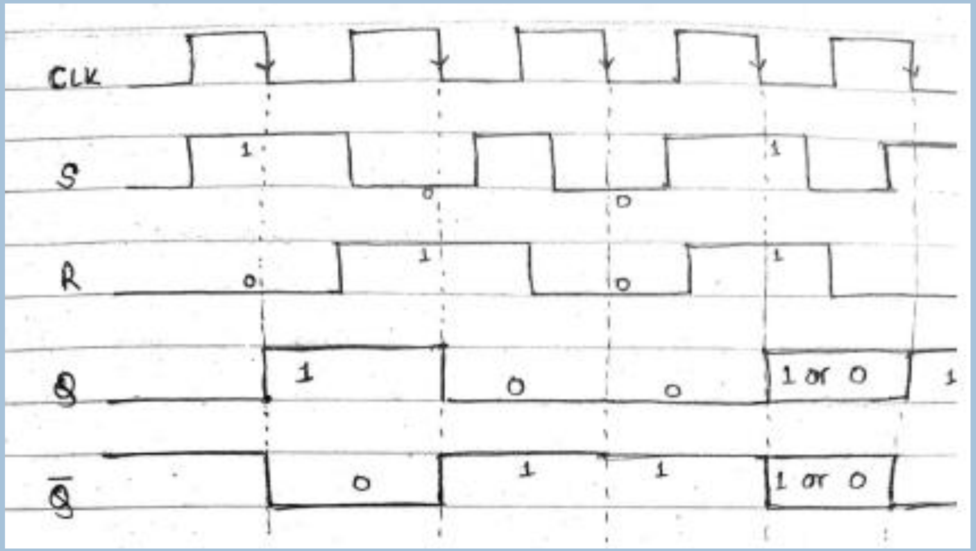


$$Q\,(t+1) = S + R'Q\,(t)$$

K-MAP and Boolean function

SR Flip-flop			
Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Excitation Table

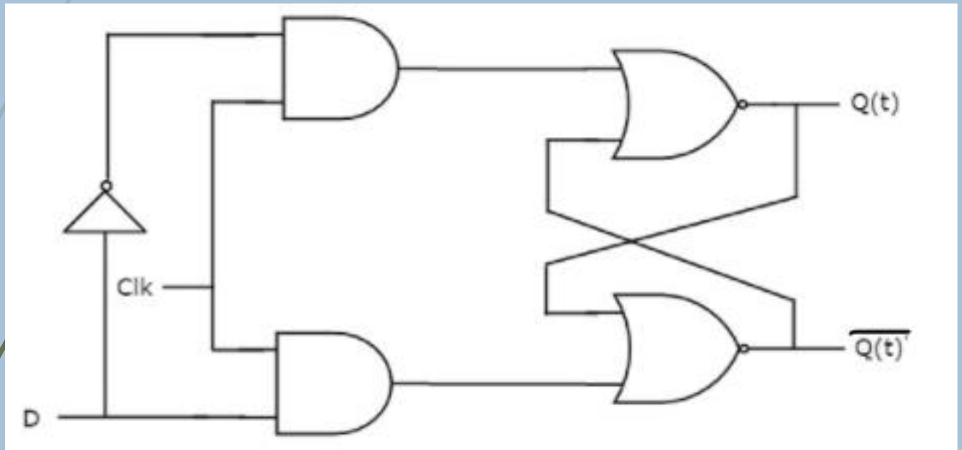


Timing diagram

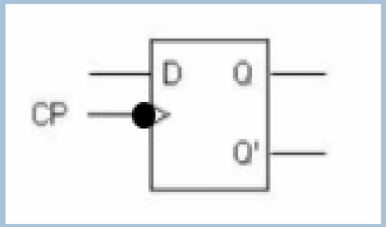
FLIP-FLOP – D FLIP-FLOP

D (Data) flip-flop operates with only positive clock transitions or negative clock transitions. Whereas, D latch operates with enable signal.

- That means, the output of D flip-flop is insensitive to the changes in the input, D except for active transition of the clock signal. The circuit diagram of D flip-flop is shown in the following figure.



Logic diagram



Graphical Symbol

Clock	D	Q(t)	Q(t+1)	Comment
↓	0	0	0	Reset
↓	0	1	0	Reset
↓	1	0	1	Set
↓	1	1	1	Set

Characteristic Table

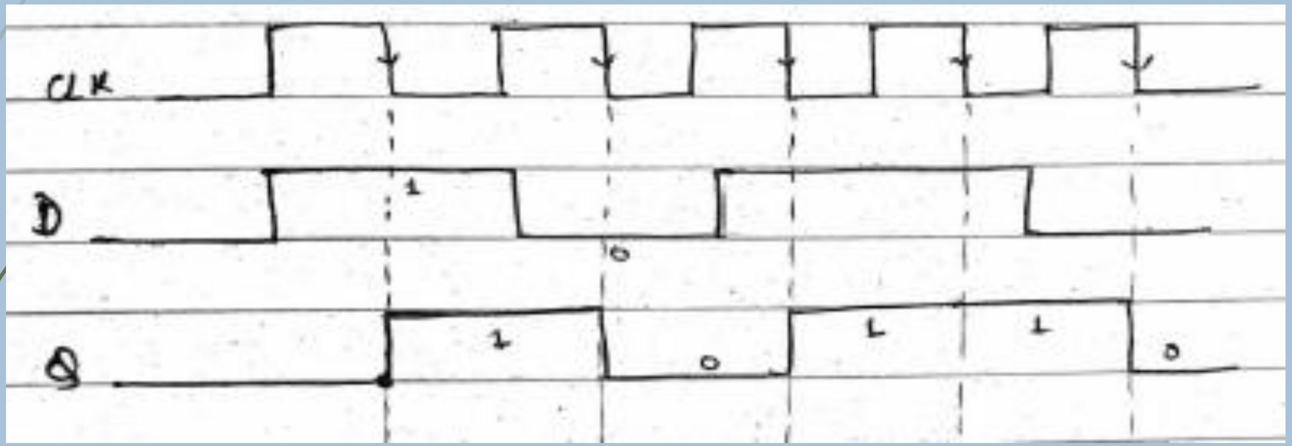
FLIP-FLOP – D FLIP-FLOP

$$Q(t+1) = D$$

Boolean function

D Flip-flop		
Q(t)	Q(t+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table

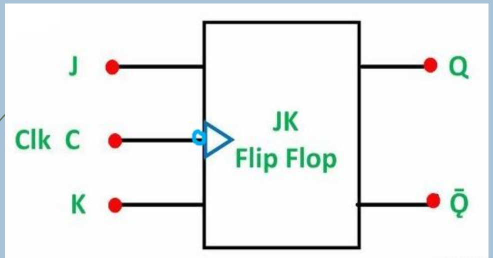


Timing diagram

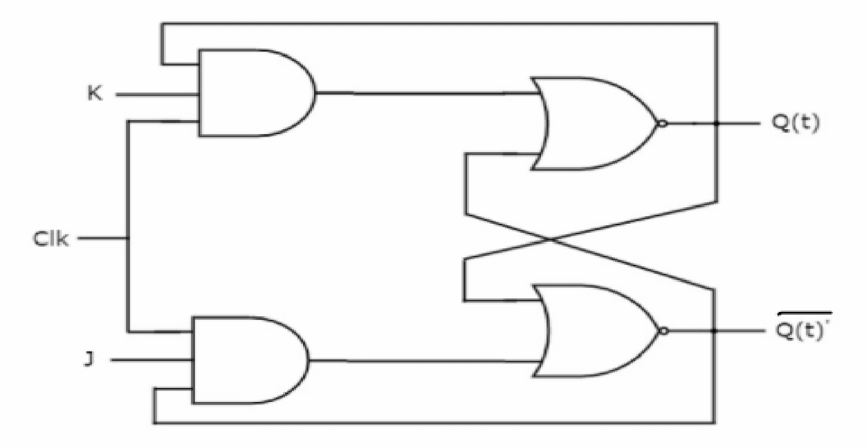
FLIP-FLOP – JK FLIP-FLOP

JK flip-flop is the modified version of SR flip-flop.

- It operates with only positive clock transitions or negative clock transitions. The circuit diagram of JK flip-flop is shown in the following figure.
- The indeterminant state of RS flip-flop is defined in JK flip-flop.



Graphical Symbol

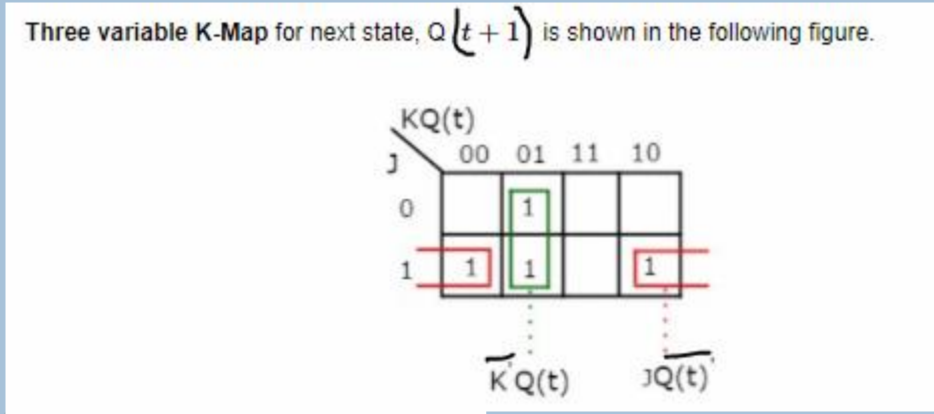


Logic diagram

Clock	J	K	Q(t)	Q(t+1)	Comment
↓	0	0	0	0	Hold
↓	0	0	1	1	Hold
↓	0	1	0	0	Reset
↓	0	1	1	0	Reset
↓	1	0	0	1	Set
↓	1	0	1	1	Set
↓	1	1	0	1	Complement
↓	1	1	1	0	Complement

Characteristic Table

FLIP-FLOP – JK FLIP-FLOP

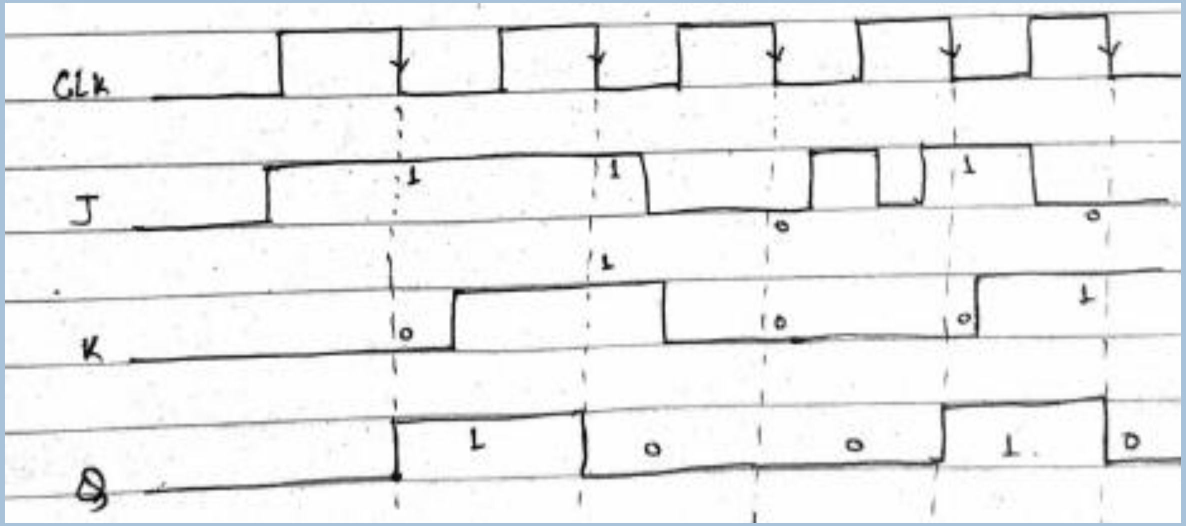


$$Q(t+1) = JQ(t)' + K'Q(t)$$

K-MAP and Boolean function

JK flip-flop			
Q(t)	Q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Excitation Table

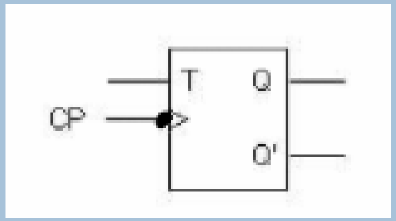


Timing diagram

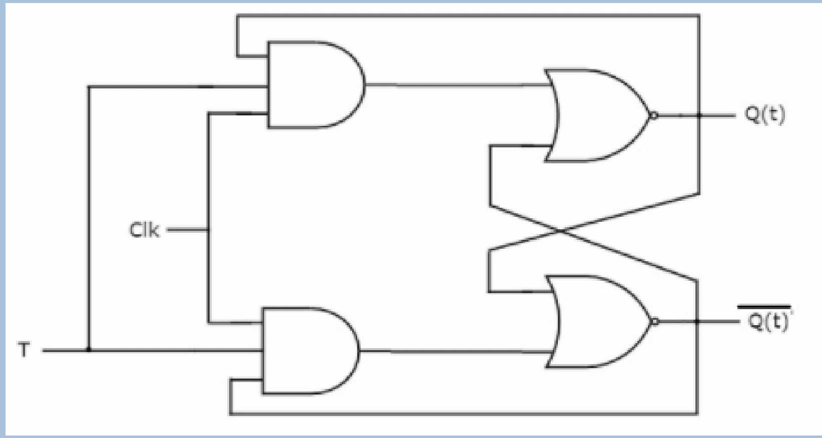
FLIP-FLOP – T FLIP-FLOP

The T flipflop is a single input version of the JK flipflop.

- The T flipflop is obtained from JK type if both inputs are tied together.
- The designation T comes from the ability of the flipflop to "Toggle", or change state. Regardless of the present state of the flipflop, it assumes the complement state when the clock pulse occurs while input T is in logic 1.



Graphical Symbol



Logic diagram

Clock	T	Q(t)	Q(t+1)	Comment
↓	0	0	0	Hold
↓	0	1	1	Hold
↓	1	0	1	Complement
↓	1	1	0	Complement

Characteristic Table

FLIP-FLOP – T FLIP-FLOP

From the above characteristic table, we can directly write the **next state equation** as

$$Q\,(t+1) = T'Q\,(t) + TQ(t)'$$

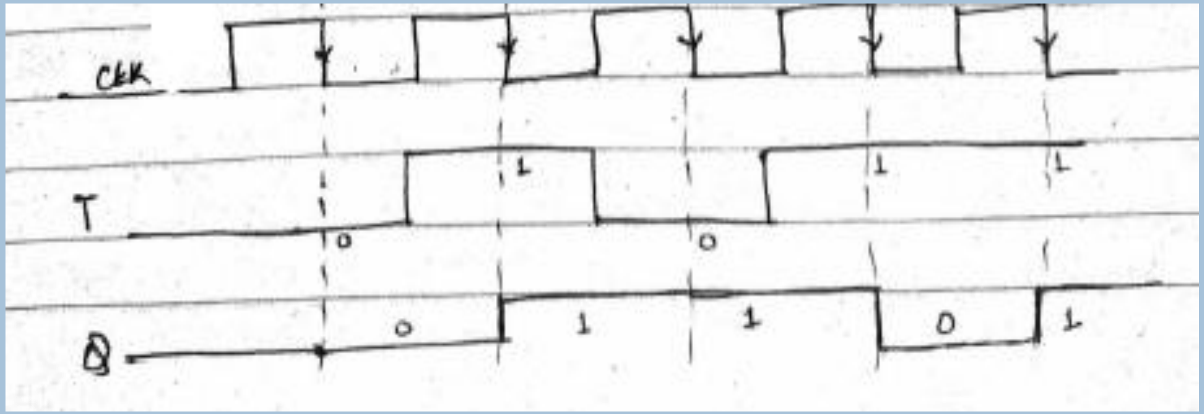
$$\Rightarrow Q\,(t+1) = T \oplus Q\,(t)$$

Boolean function

T flip-flop

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table



Timing diagram

MASTER – SLAVE JK FLIPFLOP

Race Around Condition In JK Flip-flop

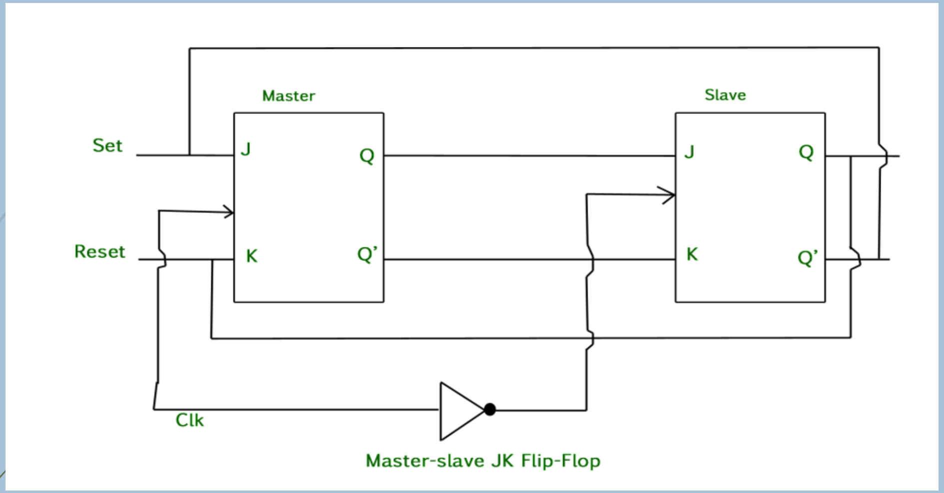
- For J-K flip-flop, if J=K=1, and if clk=1 for a long period of time, then Q output will toggle as long as CLK is high, which makes the output of the flip-flop unstable or uncertain.
- This problem is called race around condition in J-K flip-flop.
- This problem (Race Around Condition) can be avoided by ensuring that the clock input is at logic “1” only for a very short time.
- To eliminate race around condition:
  - Use edge- triggered flip-flop
  - Use **Master Slave JK flip flop**.

MASTER – SLAVE JK FLIPFLOP

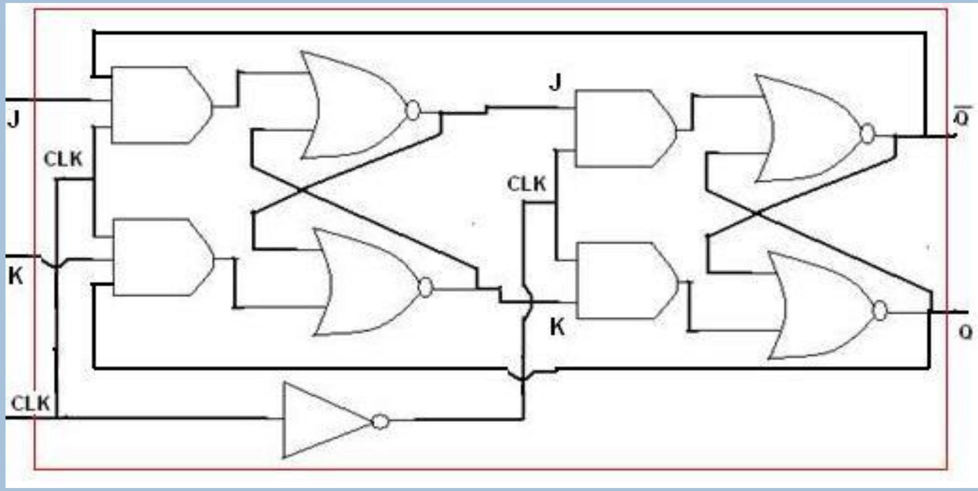
Master Slave JK flip flop

- The Master-Slave Flip-Flop is basically a combination of two JK flip-flops connected together in a series configuration.
- Out of these, one acts as the “master” and the other as a “slave”.
- The output from the master flip flop is connected to the two inputs of the slave flip flop whose output is fed back to inputs of the master flip flop.
- In addition to these two flip-flops, the circuit also includes an inverter.
- The inverter is connected to clock pulse in such a way that the inverted clock pulse is given to the slave flip-flop.
- In other words if CP=0 for a master flip-flop, then CP=1 for a slave flip-flop and if CP=1 for master flip flop then it becomes 0 for slave flip flop.

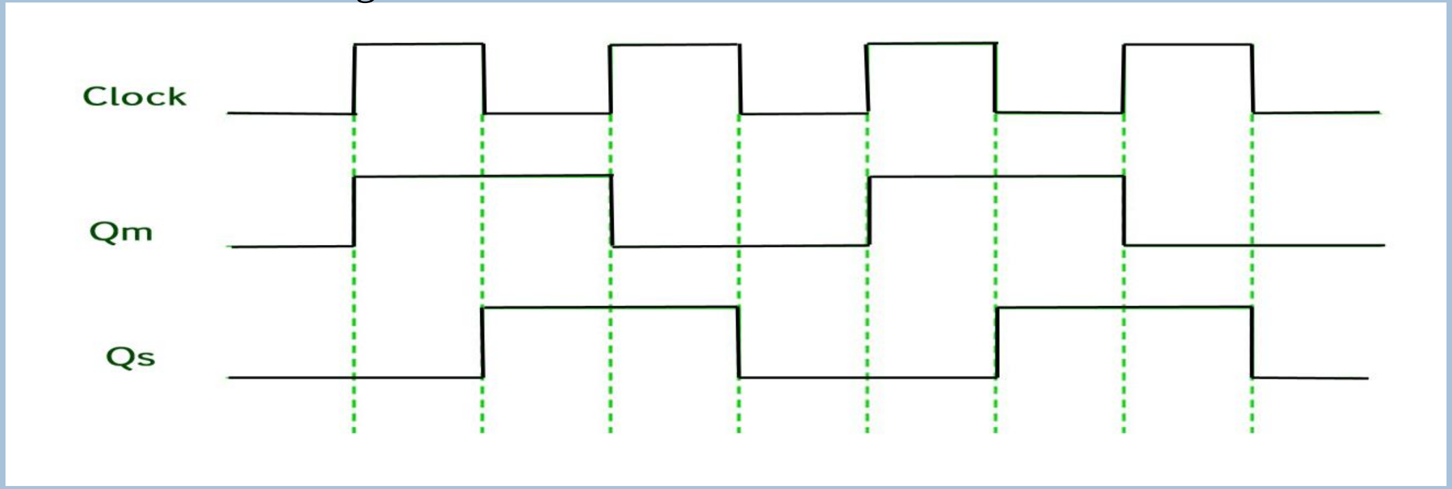
MASTER – SLAVE JK FLIPFLOP



Block diagram



Logic diagram



Timing diagram

COUNTERS

Counter is a sequential circuit.

- A digital circuit which is used for a counting pulses is known counter.
- Counter is the widest application of flip-flops. It is a group of flip-flops with a clock signal applied.
- Counters are of two types.
- **Asynchronous or ripple counters.**
  - In asynchronous counter we don't use universal clock, only first flip flop is driven by main clock and the clock input of rest of the following flip flop is driven by output of previous flip flops
- **Synchronous counters.**
  - Unlike the asynchronous counter, synchronous counter has one global clock which drives each flip flop so output changes in parallel.
  - The one advantage of synchronous counter over asynchronous counter is, it can operate on higher frequency than asynchronous counter as it does not have cumulative delay because of same clock is given to each flip flop

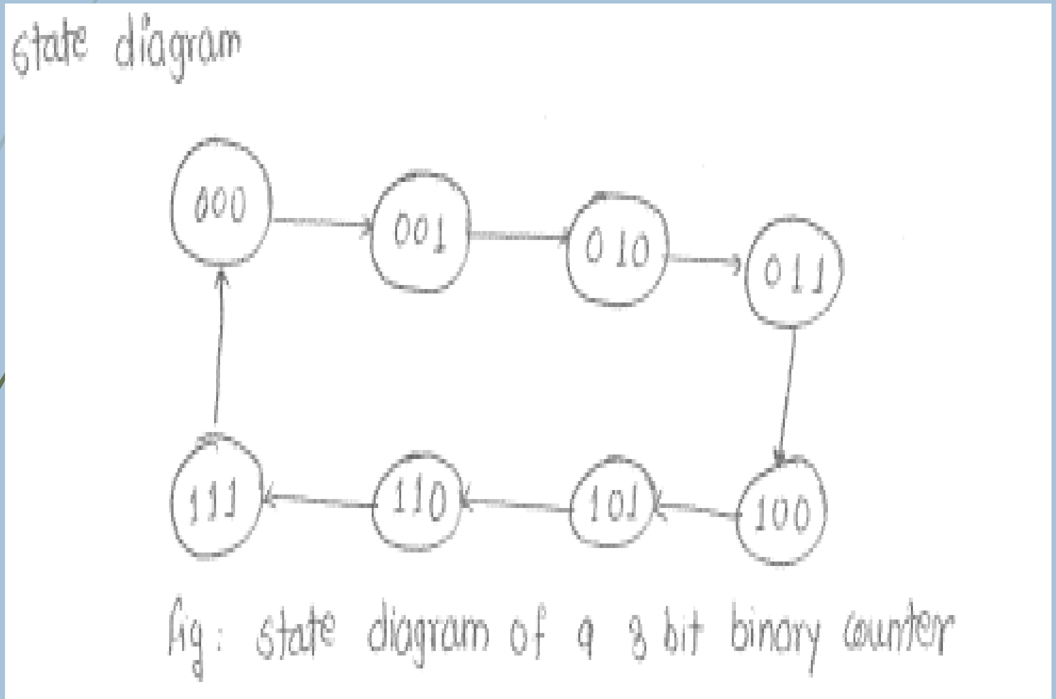
SYNCHRONOUS VS ASYNCHRONOUS COUNTERS

Sr. No.	Parameter	Asynchronous counter	Synchronous counter
1.	Circuit complexity	Logic circuit is simple.	With increase in number of states, the logic circuit becomes complicated.
2.	Connection pattern	Output of the preceding FF, is connected to clock of the next FF.	There is no connection between output of preceding FF and CLK of next one.
3.	Clock input	All the FFs are not clocked simultaneously.	All FFs receive clock signal simultaneously.
4.	Propagation delay	P.D. = n * (td) where n is number of FF and td is p.d. per FF.	P.D. = n * (td)FF + (td)gate. It is much shorter than that of asynchronous counter.
5.	Maximum frequency of operation	Low because of the long propagation delay.	High due to shorter propagation delay.

ASYNCHRONOUS COUNTERS

BINARY UP COUNTER

- An 'N' bit Asynchronous binary up counter consists of 'N' T flip-flops. It counts from 0 to $2^N - 1$.
- **EXAMPLE: 3-BIT BINARY UP COUNTER**



State diagram

No of negative edge of Clock	Q_2 *MSB*	Q_1	Q_0 *LSB*
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Count Sequence

REGISTERS

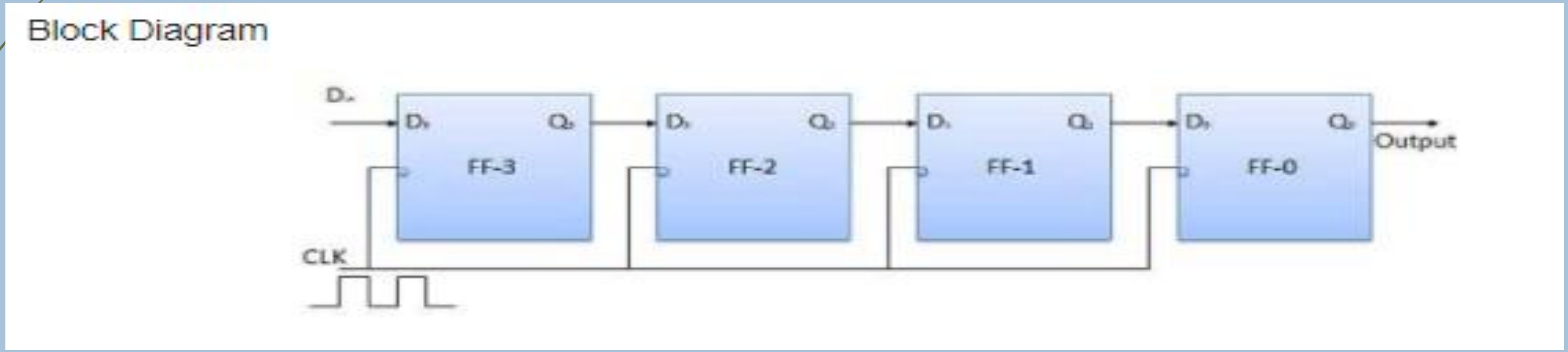
Flip-flop is a 1 bit memory cell which can be used for storing the digital data.

- To increase the storage capacity in terms of number of bits, we have to use a group of flip-flop.
- Such a group of flip-flop is known as a Register.
- The n-bit register will consist of n number of flip-flop and it is capable of storing an n-bit word.
- The binary data in a register can be moved within the register from one flip-flop to another.
- The registers that allow such data transfers are called as shift registers. There are four mode of operations of a shift register.
  - Serial Input Serial Output
  - Serial Input Parallel Output
  - Parallel Input Serial Output
  - Parallel Input Parallel Output

REGISTERS

SERIAL INPUT SERIAL OUTPUT

- Let all the flip-flop be initially in the reset condition i.e. Q3 = Q2 = Q1 = Q0 = 0. If an entry of a four bit binary number 1 1 1 1 is made into the register, this number should be applied to Din bit with the LSB bit applied first. The D input of FF-3 i.e. D3 is connected to serial data input Din. Output of FF-3 i.e. Q3 is connected to the input of the next flip-flop i.e. D2 and so on.
- Also known as, shift right register.



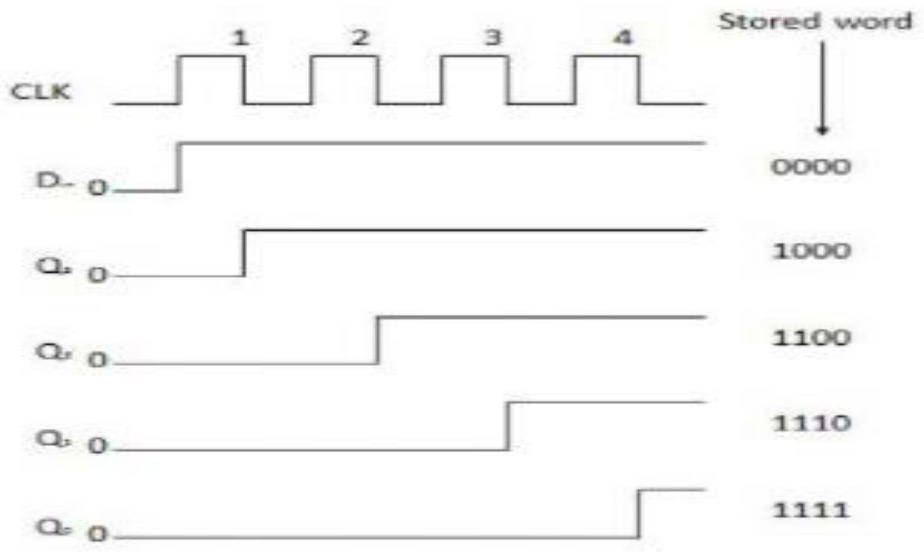
REGISTERS

Truth Table

	CLK	$D_{in} = Q_3$	$Q_3 = D_2$	$Q_2 = D_1$	$Q_1 = D_0$	Q_0
Initially			0	0	0	0
(i)	↓	1	1	0	0	0
(ii)	↓	1	1	1	0	0
(iii)	↓	1	1	1	1	0
(iv)	↓	1	1	1	1	1

⟶ Direction of data travel

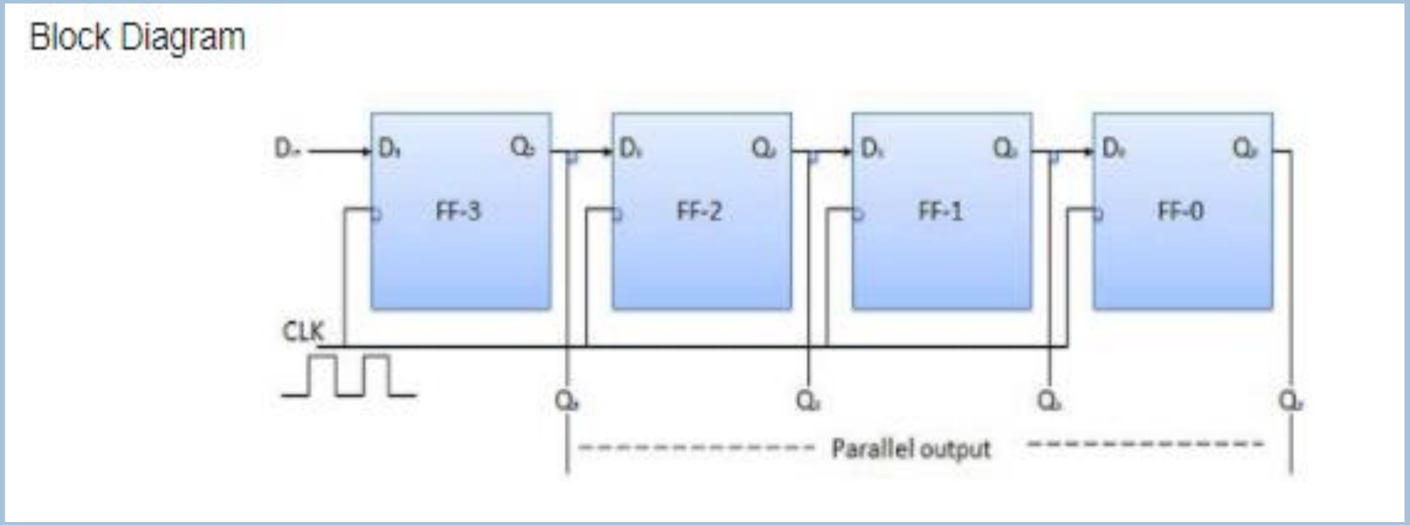
Waveforms



REGISTERS

SERIAL INPUT PARALLEL OUTPUT

- In such types of operations, the data is entered serially and taken out in parallel fashion.
- Data is loaded bit by bit. The outputs are disabled as long as the data is loading.
- As soon as the data loading gets completed, all the flip-flops contain their required data, the outputs are enabled so that all the loaded data is made available over all the output lines at the same time.
- 4 clock cycles are required to load a four bit word. Hence the speed of operation of SIPO mode is same as that of SISO mode.

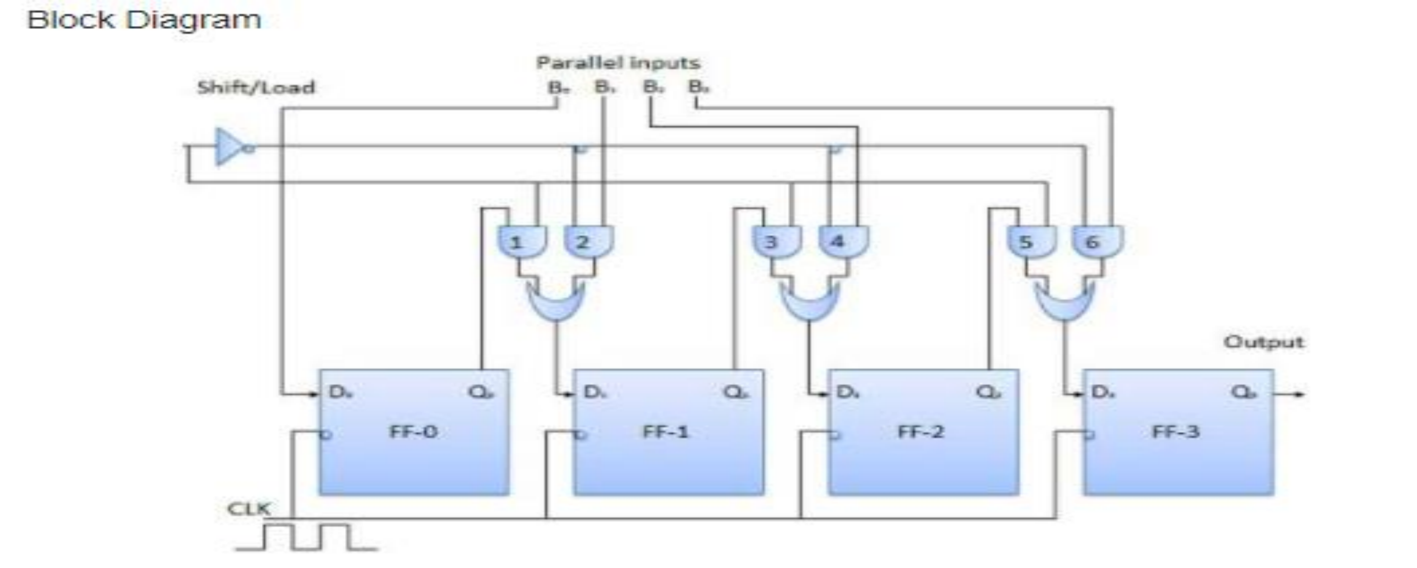


REGISTERS

32

PARALLEL INPUT SERIAL OUTPUT (PISO)

- Data bits are entered in parallel fashion.
- The circuit shown below is a four bit parallel input serial output register.
- Output of previous Flip Flop is connected to the input of the next one via a combinational circuit.
- The binary input word B0, B1, B2, B3 is applied though the same combinational circuit.
- There are two modes in which this circuit can work namely - shift mode or load mode.



REGISTERS

Load mode

- When the shift/load bar line is low (0), the AND gate 2, 4 and 6 become active they will pass B1, B2, B3 bits to the corresponding flip-flops. On the low going edge of clock, the binary input B0, B1, B2, B3 will get loaded into the corresponding flip-flops. Thus parallel loading takes place.

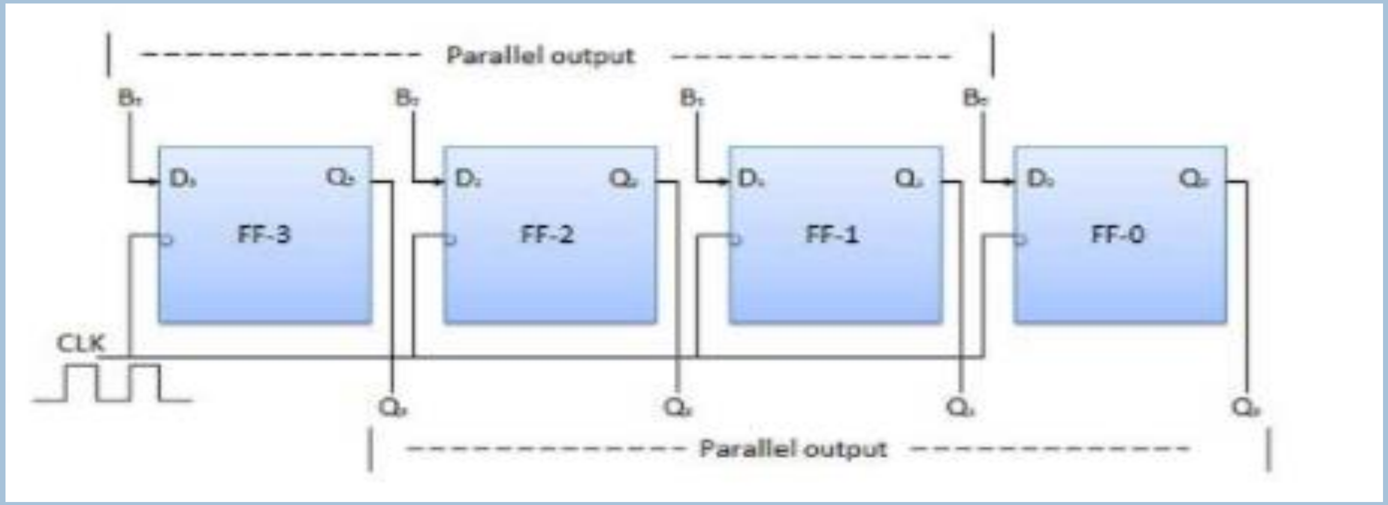
- ## Shift mode

  - When the shift/load bar line is high (1), the AND gate 2, 4 and 6 become inactive. Hence the parallel loading of the data becomes impossible. But the AND gate 1,3 and 5 become active. Therefore the shifting of data from left to right bit by bit on application of clock pulses. Thus the parallel in serial out operation takes place.

REGISTERS

PARALLEL INPUT PARALLEL OUTPUT (PIPO)

- In this mode, the 4 bit binary input B0, B1, B2, B3 is applied to the data inputs D0, D1, D2, D3 respectively of the four flip-flops.
- As soon as a negative clock edge is applied, the input binary bits will be loaded into the flip-flops simultaneously. The loaded bits will appear simultaneously to the output side. Only clock pulse is essential to load all the bits.



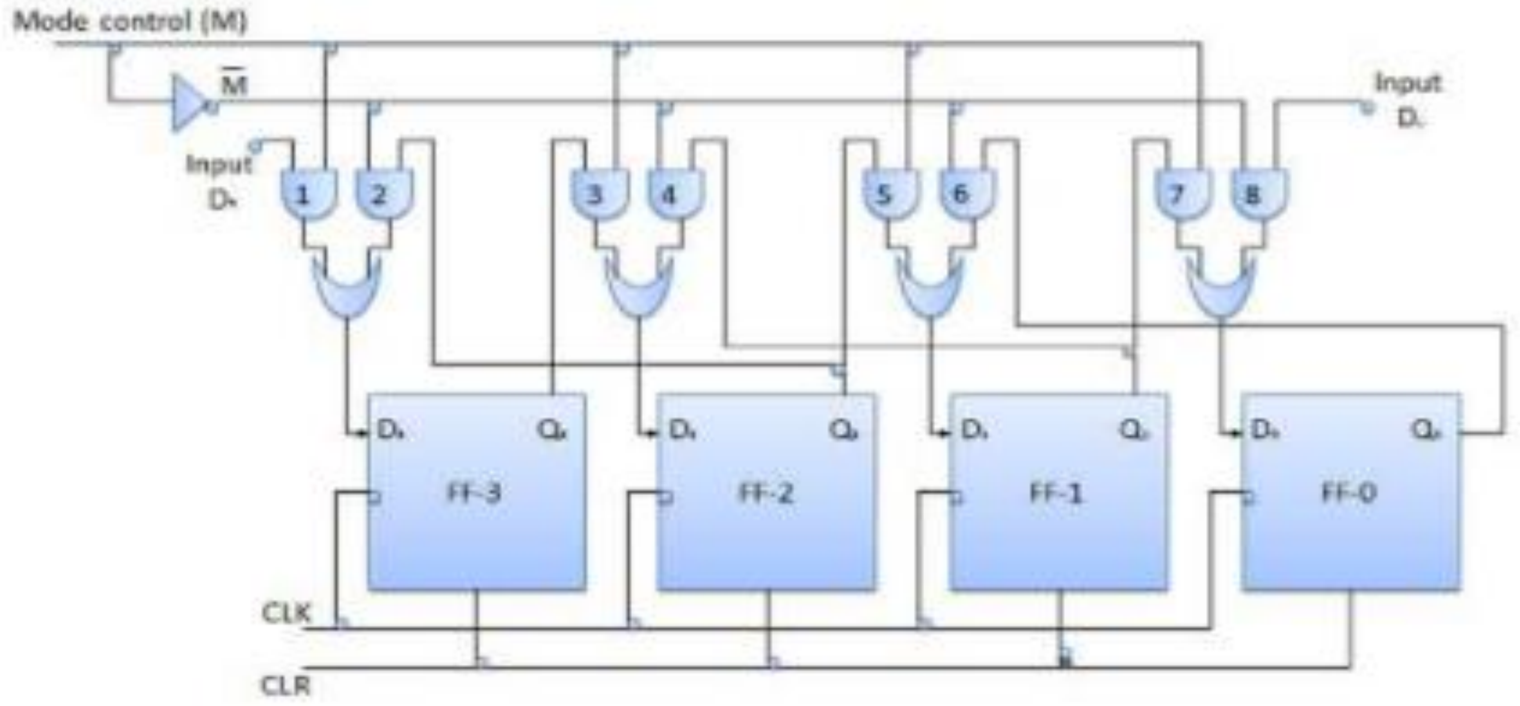
SHIFT REGISTERS

Bidirectional Shift Register

- If a binary number is shifted left by one position then it is equivalent to multiplying the original number by 2. Similarly if a binary number is shifted right by one position then it is equivalent to dividing the original number by 2.
- Hence if we want to use the shift register to multiply and divide the given binary number, then we should be able to move the data in either left or right direction.
- Such a register is called bi-directional register. A four bit bi-directional shift register is shown in fig.
- There are two serial inputs namely the serial right shift data input DR, and the serial left shift data input DL along with a mode select input (M).

SHIFT REGISTERS

Block Diagram



SHIFT REGISTERS

With M = 1 – Shift right operation

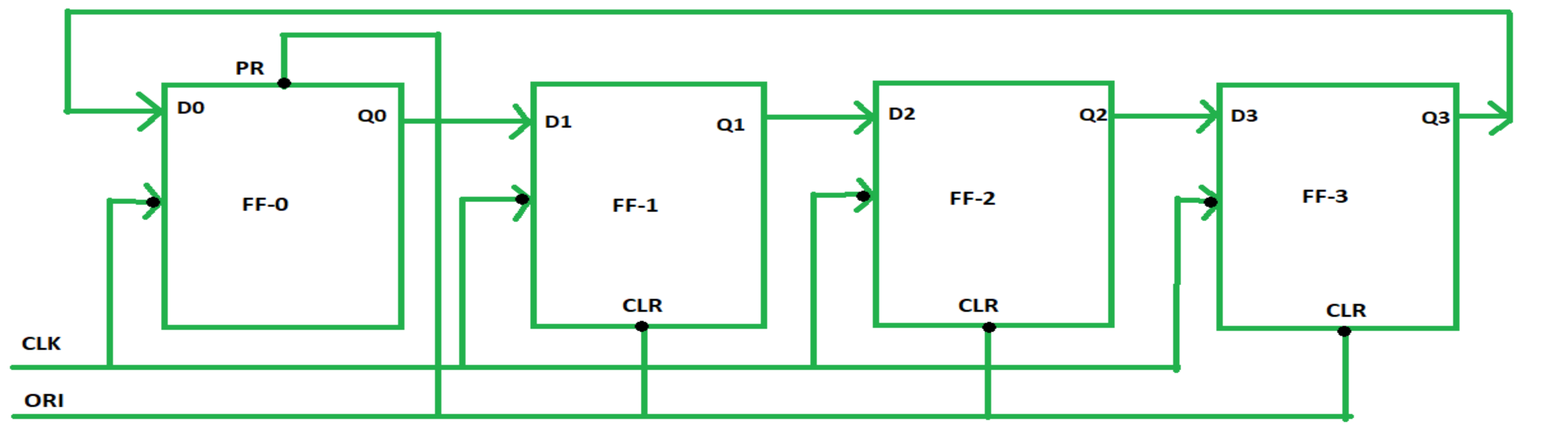
- If M = 1, then the AND gates 1, 3, 5 and 7 are enabled whereas the remaining AND gates 2, 4, 6 and 8 will be disabled.
- The data at DR is shifted to right bit by bit from FF-3 to FF-0 on the application of clock pulses. Thus with M = 1 we get the serial right shift operation

With M = 0 – Shift left operation

- When the mode control M is connected to 0 then the AND gates 2, 4, 6 and 8 are enabled while 1, 3, 5 and 7 are disabled.
- The data at DL is shifted left bit by bit from FF-0 to FF-3 on the application of clock pulses. Thus with M = 0 we get the serial right shift operation.

RING COUNTERS

- Ring counter is a typical application of Shift resister.
- Ring counter is almost same as the shift counter.
- The only change is that the output of the last flip-flop is connected to the input of the first flip-flop in case of ring counter but in case of shift resister it is taken as output. Except this all the other things are same.



Ring Counter

- In this diagram, we can see that the clock pulse (CLK) is applied to all the flip-flop simultaneously. Therefore, it is a Synchronous Counter.

RING COUNTERS

Also, here we use Overriding input (ORI) to each flip-flop. Preset (PR) and Clear (CLR) are used as ORI.

- When PR is 0, then the output is 1. And when CLR is 0, then the output is 0. Both PR and CLR are active low signal that is always works in value 0.

PRESETED 1

ORI	CLK	Q0	Q1	Q2	Q3
low	X	1	0	0	0
high	low	0	1	0	0
high	low	0	0	1	0
high	low	0	0	0	1
high	low	1	0	0	0

RING COUNTERS

This Preseted 1 is generated by making ORI low and that time Clock (CLK) becomes don't care.

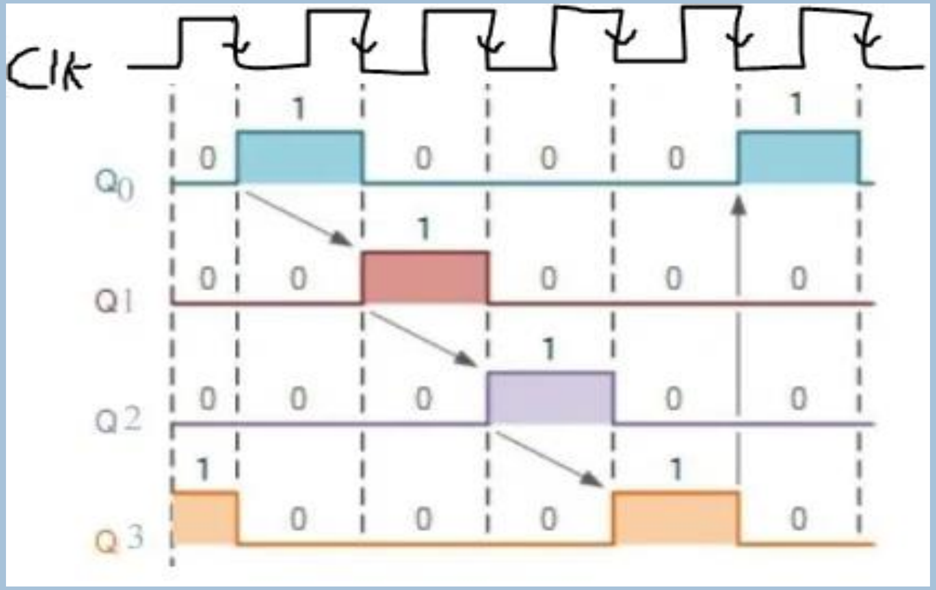
- After that ORI made to high and apply low clock pulse signal as the Clock (CLK) is negative edge triggered.
- After that, at each clock pulse the preseted 1 is shifted to the next flip-flop and thus form Ring.
- From the above table, we can say that there are 4 states in 4-bit Ring Counter.
- 4 states are:

1 0 0 0

0 1 0 0

0 0 1 0

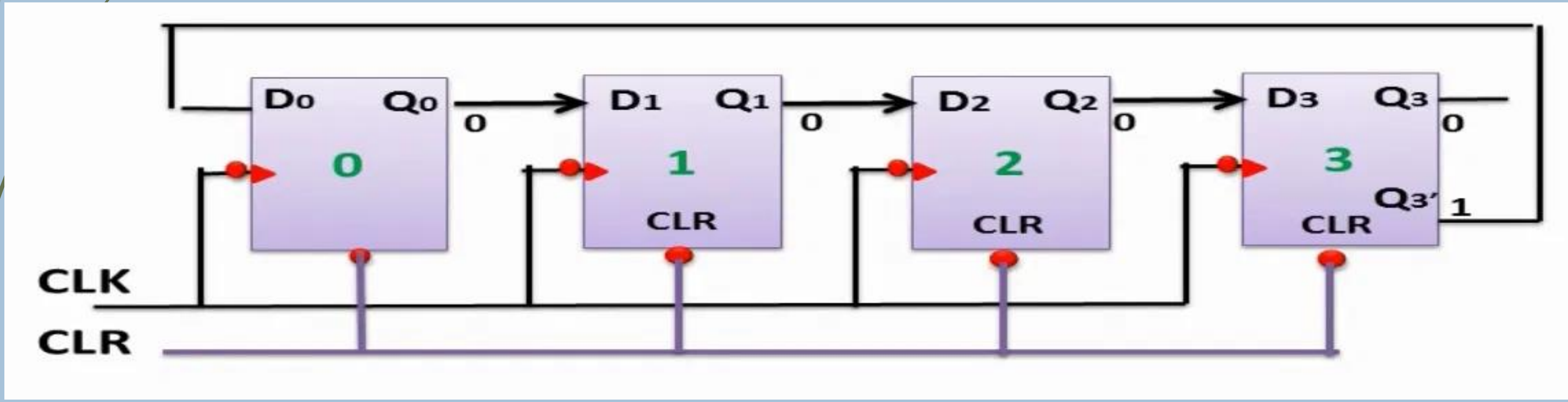
0 0 0 1



JOHNSON COUNTERS

In the ring counter we given the output of the last flip flop into the input of the first flip but in the Johnson counter the last flip flop complemented output is given to the input of the first flip flop.

- In Johnson counter the number of states is equal to twice the number of flip flops.
- So if we use 4 flip flops we will have 8 states so the number of the states are double.
- We applied clock simultaneously to all flip flops.
- The clear input is applied to all the flip flops.



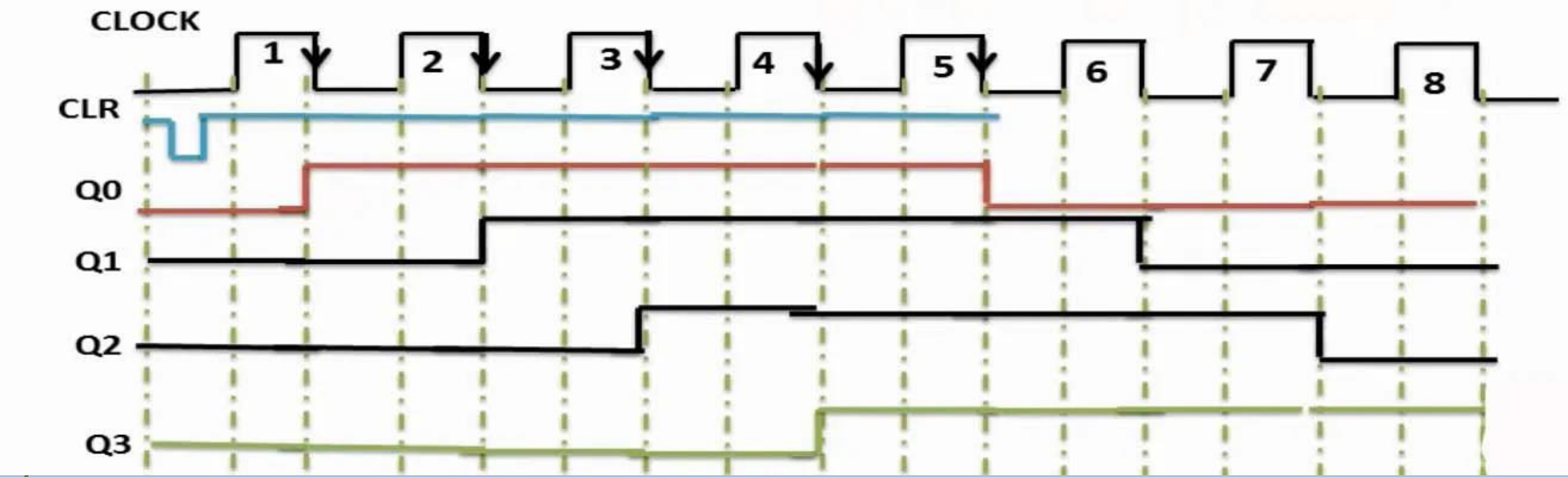
JOHNSON COUNTERS

The output of the first flip flop which is Q0 is given at the input of the second flip flop D1 and the output of the second flip flop which is Q2 is given to input of the third flip flop which is D2 and the complemented output (Q3') will be given to the input of the first flip D0.

- The difference between the ring counter and Johnson counter is that it does not require pre-set.

Clear/ Pre-set	Clock	Q0	Q1	Q2	Q3
0	No clock	0	0	0	0
1	↓	1	0	0	0
1	↓	1	1	0	0
1	↓	1	1	1	0
1	↓	1	1	1	1
1	↓	0	1	1	1
1	↓	0	0	1	1
1	↓	0	0	0	1
1	↓	0	0	0	0

JOHNSON COUNTERS



Sequential Circuit

1

. Latches constructed with NOR and NAND gates tend to remain in the latched condition due to which configuration feature?

a) Low input voltages

b) Synchronous operation

c) Gate impedance

d) Cross coupling

2

. When both inputs of a J-K flip-flop cycle, the output will __________

a) Be invalid

b) Change

c) Not change

d) Toggle

3

A basic S-R flip-flop can be constructed by cross-coupling of which basic logic gates?

a) AND or OR gates

b) XOR or XNOR gates

c) NOR or NAND gates

d) AND or NOR gates

Sequential Circuit

4

. The truth table for an S-R flip-flop has how many VALID entries?

a) 1

b) 2

c) 3

d) 4

5

Which of the following is correct for a gated D-type flip-flop?

a) The Q output is either SET or RESET as soon as the D input goes HIGH or LOW

b) The output complement follows the input when enabled

c) Only one of the inputs can be HIGH at a time

d) The output toggles if one of the inputs is held HIGH

6

. The logic circuits whose outputs at any instant of time depends only on the present input but also on the past outputs are called ____________

a) Combinational circuits

b) Sequential circuits

c) Latches

d) Flip-flops

Sequential Circuit

7. Whose operations are more faster among the following?

a) Combinational circuits
b) Sequential circuits
c) Latches
d) Flip-flops

8. In S-R flip-flop, if Q = 0 the output is said to be ___________

a) Set
b) Reset
c) Previous state
d) Current state

9. What is a trigger pulse?

a) A pulse that starts a cycle of operation
b) A pulse that reverses the cycle of operation
c) A pulse that prevents a cycle of operation
d) A pulse that enhances a cycle of operation

Sequential Circuit

10 . How many types of sequential circuits are?

a) 2

b) 3

c) 4

d) 5

11 . The output of latches will remain in set/reset untill ___________

a) The trigger pulse is given to change the state

b) Any pulse given to go into previous state

c) They don't get any pulse more

d) The pulse is edge-triggered

12 The data sheet of a certain flip-flop specifies that the minimum HIGH time $t_W(H)$ for the clock pulse is 16 nanoseconds and the minimum LOW time $t_W(L)$ is 29 nanoseconds. What is the maximum operating frequency for the given flip-flop?

1. 62·50 MHz
2. 31·25 MHz
3. 22·22 MHz
4. 11·11 MHz

Sequential Circuit

13 In which flip flop the present input will be the next output?

- S-R
- J-K
- D
- T

14 The clear input is used to make output _____

- Q=1
- Q=0
- Invalid
- No change

15 A flip flop is an _________

- Edge sensitive device
- Synchronous device
- Both a and b
- None of the above

Sequential Circuit

16 The preset input is used to make output _____

- Q=1
- Q=0
- Invalid
- No change

17 The shift registers are categorized into ______

- One
- Two
- Three
- Four

18 _______ are the applications of flip flop

- Registers
- Counters
- Storage devices
- All of the above

Sequential Circuit

19 **The number of flip-flops in a shift register is dependent upon?**

- ○ The modulus of the counter
- ○ The number of stages in the counter
- ○ The number of digits to be stored
- ○ None of the above

20 **. A shift register is a digital circuit that ___________**

- ○ Stores data.
- ○ Shifts the data from left to right.
- ○ Shifts the data from right to left.
- ○ all of the above.

21 **What type of register is a shift register?**

- ○ Digital
- ○ Analog
- ○ Both 1 and 2
- ○ Neither 1 nor 2

Sequential Circuit

22

. In which mode can we provide data to all the flip-flops simultaneously?

- Loopback mode
- Serial input mode
- Parallel input mode
- All of the above

23

. What is a shift register?

- An adder circuit
- A memory circuit
- A combinational circuit
- A decoder circuit

24

Which of the following is true about shift registers?

- It is not used to store multi-bit data.
- They are available only in the parallel mode of operation.
- They are useful for data transfer from one location to another.
- All of the above.

Sequential Circuit

25

Based on how binary information is entered or shifted out, shift registers are classified into ______ categories.

a) 2
b) 3
c) 4
d) 5

26

A shift register that will accept a parallel input or a bidirectional serial load and internal shift features is called as?

a) Tristate
b) End around
c) Universal
d) Conversion

27

The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output shift register with an initial state 01110. After three clock pulses, the register contains _______

a) 01110
b) 00001
c) 00101
d) 00110

Sequential Circuit

28 The full form of SIPO is ___________

a) Serial-in Parallel-out

b) Parallel-in Serial-out

c) Serial-in Serial-out

d) Serial-In Peripheral-Out

29 . How can parallel data be taken out of a shift register simultaneously?

a) Use the Q output of the first FF

b) Use the Q output of the last FF

c) Tie all of the Q outputs together

d) Use the Q output of each FF

30 . Assume that a 4-bit serial in/serial out shift register is initially clear. We wish to store the nibble 1100. What will be the 4-bit pattern after the second clock pulse? (Right-most bit first)

a) 1100

b) 0011

c) 0000

d) 1111

Sequential Circuit

31 A serial in/parallel out, 4-bit shift register initially contains all 1s. The data nibble 0111 is waiting to enter. After four clock pulses, the register contains ________

a) 0000

b) 1111

c) 0111

d) 1000

32 How many flip-flops are required to make a MOD-32 binary counter?

(A) 3

(B) 45

(C) 5

(D) 6

33 A MOD-16 ripple counter is holding the count 1001_2. What will the count be after 31 clock pulses?

(A) 1000_2

(B) 1010_2

(C) 1011_2

(D) 1101_2

Sequential Circuit

34 With a 200 kHz clock frequency, eight bits can be serially entered into a shift register in ______

a) 4 μs

b) 40 μs

c) 400 μs

d) 40 ms

35 Using four cascaded counters with a total of 16 bits, how many states must be deleted to achieve a modulus of 50,000?

- (A) 50,000
- (B) 65,536
- (C) 25,536
- (D) 15,536

36 The terminal count of a modulus-11 binary counter is ________.

- (A) 1010
- (B) 1000
- (C) 1001
- (D) 1100

Sequential Circuit

37

An eight-stage ripple counter uses a flip-flop with propagation delay of 75 nanoseconds. The pulse width of the strobe is 50ns. Frequency of the input signal which can be used for proper operation of the counter is approximately

2 points

- a. 1MHz
- b. 2MHz
- c. 500MHz
- d. 4MHz

Concept:

In case of ripple or asynchronous counter total propagation delay $T_{pd} = n\ t_{pd}$

Where n = no. of flip flops

t_{pd} = propagation delay of each flip flop

If the strobe signal is given, the pulse width of the strobe is also added, i.e.

$T_{pd} = n\ t_{pd} + T_s$

Sequential Circuit

38 A 4-bit ripple counter consists of flip-flop that each have propagations delay form clock to Q output of 12ns. For the counter to recycle from 1111 to 0000, it takes a total of

2 points

- a. 12ns
- b. 48ns
- c. 24ns
- d. 36ns

Concept:

- For an n-bit ripple counter, the MSB is generated only when the carry from all the previous flip-flip is propagated to the MSB flip flop.
- So, the maximum time(Worst-Case delay) taken for the output of the **Ripple counter to be stable = n × t_d** (where t_d is the propagation delay of each flip flop)

Sequential Circuit

39 The divide by 60 counter in digital clock is implemented by using two cascading counters 2 points

- Mod-6, Mod-10
- Mod-50, Mod-10
- Mod-10. Mod-50
- Mod-50, Mod-6

40 In a certain digital waveform, the period is twice the pulse width. the duty cycle is ___ % 2 points

- 100
- 50
- 200
- 25

Sequential Circuit

41 A flip-flop circuit can be used for:

1. counting
2. scaling
3. demodulation
4. More than one of the above

42 A flip-flop can store:

1. One bit of data
2. Two bits of data
3. Three bits of data
4. Any number of bits of data

43 A single flip-flop is a modulo ________ counter.

1. 1
2. 2
3. 0
4. More than one of the above

44 Race Around condition can be avoided in Digital logic circuits using?

1. Shift Register
2. Master-Slave JK Flip Flop
3. Full Adder
4. More than one of the above

Sequential Circuit

45 Which condition is shown in J-K flip flop as no changes next state from the present state?

1. J = 0, K = 0
2. J = 0, K = 1
3. J = 1, K = 0
4. J = 1, K = 1

46 D flip flop can be made from a J-K flip flop by making

1. J = K
2. J = K = 1
3. J = 0, K = 1
4. $J = \bar{K}$

47 Which is used for storing the one-bit digital data?

1. NAND GATE
2. GATE
3. Flip flop
4. Register

48 Master-slave configuration is used in FF to

1. increase its clocking rate
2. reduces power dissipation
3. eliminates race around condition
4. improves its reliability

Sequential Circuit

49 When two asynchronous active low inputs PRESET and CLEAR are applied to a J-K flip flop the output will be

1. 0
2. Undefined
3. Previous state
4. 1

50 Which of the following logic circuits do not have no-change condition?

1. D-FF
2. T-FF
3. JK-FF
4. SR-Latch

51 Which of the following is the other name for the D flip-flop?

1. Dual flip-flop
2. Decimal flip-flop
3. Delay flip-flop
4. Decay flip-flop

52 How many Flip flops circuits are needed to divide by 16?

1. Two
2. Four
3. Eight
4. Sixteen

Sequential Circuit

53 The one input RS flip flop is the ______ flip flop

1. T
2. D
3. R
4. Latch

54 If input to T flip flop is 200 Hz signal, then what will be the output signal frequency if four T flip flops are connected in cascade

1. 200 Hz
2. 50 Hz
3. 800 Hz
4. None of the above

$$f_{out} = \frac{\text{Input Frequency}}{2^n}$$

55 When both the inputs of a latch are high, the output is unpredictable. What is this condition called?

1. Bistable
2. Indeterminate
3. Inactive
4. No change

56 In S-R latch (NOR), when the SET input is made high, output Q becomes:

1. 0
2. 1
3. no change
4. application not allowed

Sequential Circuit

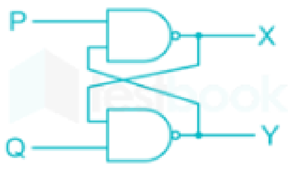
57 Which property is NOT considered in latches?

1. Output of the latches changes as we change the input.
2. changes as we change the input.
3. Latches are edge triggered.
4. Latches are fast.

58 _____ is commonly used to interface output devices.

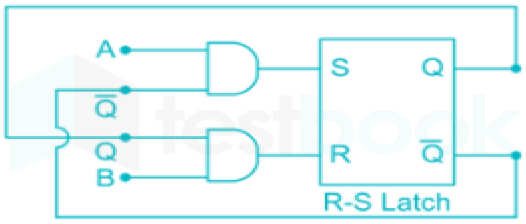
1. Buffer
2. Pulse generator
3. Accumulator
4. Latch

59 In the latch circuit shown, the NAND gates have non-zero, but unequal propagation delays. The present input conditions is: P = Q = '0'. If the input conditions is changed simultaneously to P = Q = '1', the outputs X and Y are



1. X = '1', Y = '1'
2. either X = '1', Y = '0' or X = '0', Y = '1
3. either X = '1', Y = '1' or X = '0', Y = '0'
4. X = '0', Y = '0'

60 The two inputs A and B are connected to a NOR based R-S latch, via two AND gates as shown in the figure. If A = 1 and B = 0, the output $Q\bar{Q}$ is

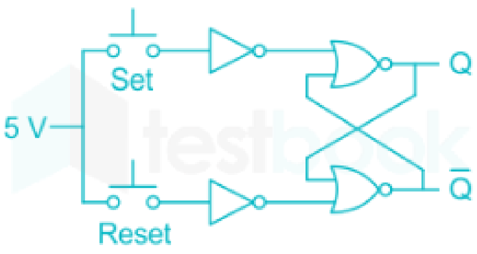


1. 00
2. 10
3. 01
4. 11

Sequential Circuit

61

An SR latch is implemented using TTL gates as shown in the figure. The set and reset pulse inputs are provided using the push-button switches. It is observed that the circuit fails to work as desired. The SR latch can be made functional by changing



1. NOR gates to NAND gates
2. inverters to buffers
3. NOR gates to NAND gates and inverters to buffers
4. 5 V to ground

62

Which of the following is true?

1. A flip-flop and latch both function same
2. flip-flop is edge triggered and latch is level triggered
3. Flip-flop is level triggered and latch is edge triggered
4. Either of flip-flop and latch is a combinational circuit

63

72). The BCD is one type of counter which is also known as ________

- Synchronous
- Asynchronous
- Parallel
- Decade

64

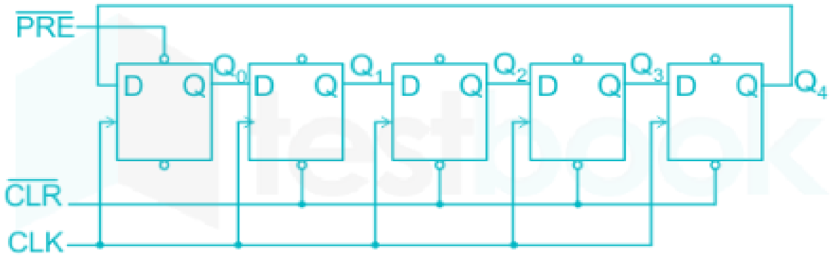
65). The flip flops are activated by __________ trigger

- Only positive edge
- Only negative edge
- Either positive or negative edge
- None of the above

Sequential Circuit

65

The following circuit is a



1. Mod-5 Ring counter
2. Mod-5 Johnson counter
3. Mod-10 Ring counter
4. Mod-10 Johnson counter

66

For the multistage counter arrangement of the given figure, determine the frequency of the output signal in Hz.

a) 125Hz
b) 1 MHz
c) 1 KHz
d) 125KHz

67

A traffic signal cycles from GREEN to YELLOW, YELLOW to RED and RED to GREEN. In each cycle, GREEN is turned on for 70 seconds, YELLOW is turned on for 5 seconds and the RED is turned on for 75 seconds. This traffic light has to be implemented using a finite state machine (FSM). The only input to this FSM is a clock of 5 second period. The minimum number of flip-flops required to implement this FSM is _______ .

a) 6
b) 30
c) 5
d) None

68

2. What is the difference between a shift-right register and a shift-left register?

a) There is no difference
b) The direction of the shift
c) Propagation delay
d) The clock input

THANK YOU